

Cyber Attack Simulation Platform

Overview

The Cyber Attack Simulation Platform is a secure cyber range environment designed to simulate cyberattacks and defensive operations for training and educational purposes.

Problem

Organizations require safe environments to train security professionals and evaluate defensive strategies without exposing production systems to risk.

Solution

The platform provides controlled attack simulations, user management, monitoring capabilities, and reporting features within an isolated environment.

Key Features

- Attack simulation scenarios
- User authentication and authorization
- Role-based access control
- Security monitoring
- Reporting and analytics
- Sandbox isolation

Methodologies Applied

- SQUARE Security Requirements Engineering
- STRIDE Threat Modeling

My Contributions

- Designed system architecture
- Developed security requirements
- Performed threat modeling
- Created system diagrams and workflows
- Designed user interfaces and prototypes

Outcome

Produced a secure training platform concept that supports cybersecurity education and hands-on attack simulation exercises.